

III Cascading Behavior in Networks

- We follow
- Why do we follow
- Diffusion in NW
- Modeling Diffusion
- impact of Communities on Diffusion
- Cascade and clusters.
 ↳ payoff ↳ same type of people

III Cascading Behavior in Social Networks :-

- ① When people are connected in Networks to each other then they can influence each other's behavior and decisions.

This is called cascading Behavior in Networks.

- ② cascade networks are an example of Networks that connect individuals on the basis of the direction in which the data or information flows between them.

Network Effects & Cascading Behaviour

- 1) Decision-based
- 2) probabilistic.

1) Decision-based :-

each node decides whether to activate based on its neighbors decisions. Deterministic rule, nodes are active players, and suited for modeling adoption.

2) Probabilistic :-

infected nodes "push" contagion to uninfected nodes with some probability.

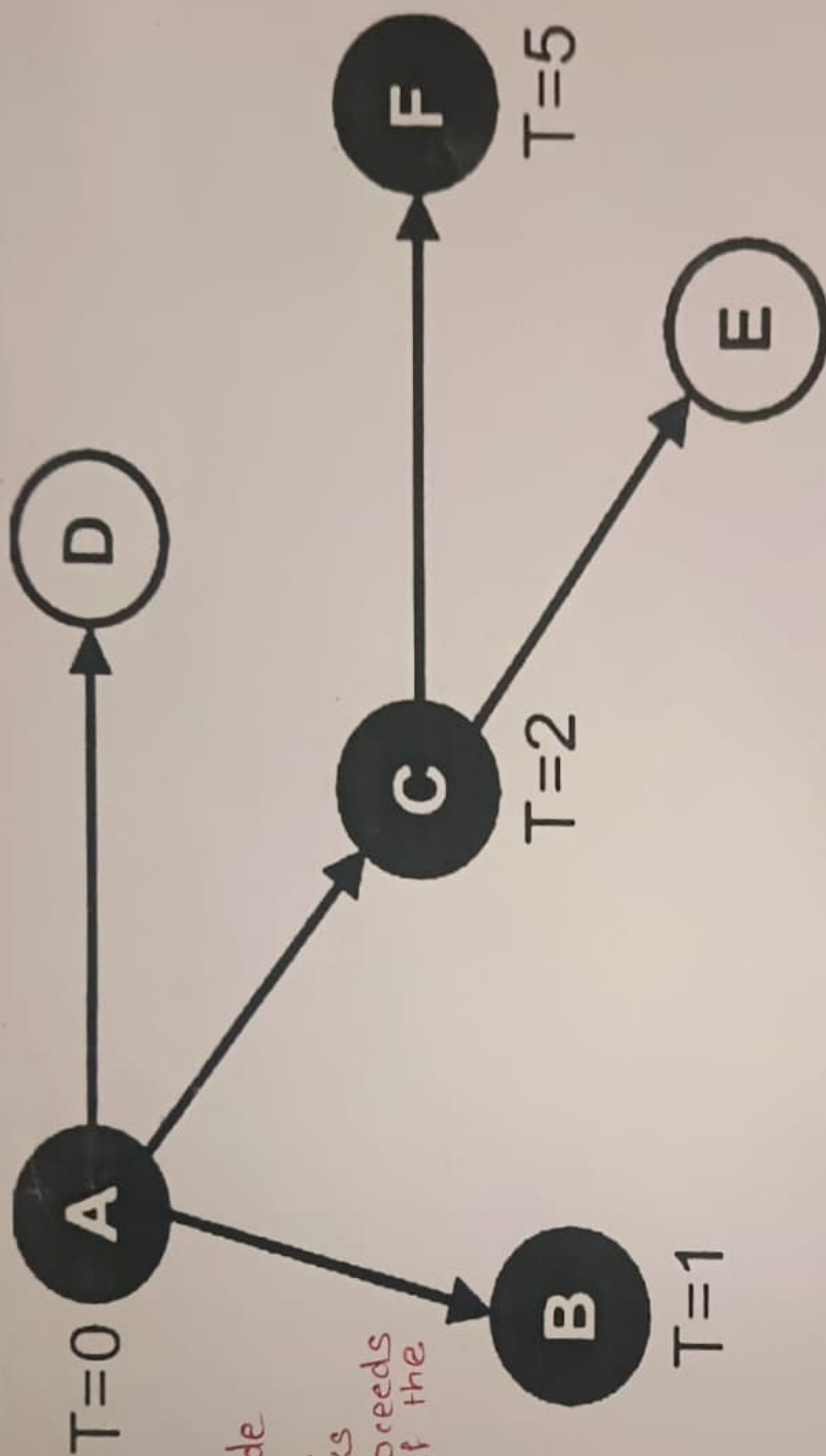
- ③ information cascades are phenomena in which individuals adopt a new action or idea due to influence by others. As such a process spreads through an underlying social NW, it can result in widespread adoption overall.

* Cluster Density :

* Clusters are obstacles to cascades

- consider an arbitrary network in which behavior A is spreading with threshold q starting from a set of initial adopters. Suppose that the remaining network contains a cluster of density greater than $1-q$. we now argue that no node inside the cluster will ever adopt A.

* information cascading example :-



- 1) ex, social cascade
- 2) network of 6 users
- 3) 5 directional links
- 4) social cascade proceeds in the direction of the arrows.

Illustrating example of a social cascade Figure 3 illustrates a social cascade in a small example network of 6 users and 5 directional links. A social cascade proceeds in the direction of the arrows. For example, a directional link from user A to B indicates that B added A as his contact. Fans of the photo are indicated by black nodes. At time $T = 0$, user A marks a photo as a favorite. Over time, users having A as a contact, namely B, C, and D, are exposed to the photo. The first step of the cascade happens when C marks the photo as a favorite at time $T = 1$. The next step of the cascade happens when B marks the photo as a favorite at $T = 2$. User D remains unaffected by the photo and is indicated by a white node. This may be because D is not interested in the photo or has not yet been exposed to it. At $T = 5$, user F discovers the new photo because he is a contact of C, marks it as a favorite, and continues the social cascade. Thus, at the end of the cascade, users A, B, C, and F are infected, while users D and E are removed. In our model, we assume that there is a set J of photos and a set U of users. Each user $u \in U$ is in one of three

Diffusion in N/W

- ① Diffusion is the process by which information is spread from one place to another through interaction.
- ② The Diffusion process involves three main elements as follows :
 - ① sender
 - ② Receiver
 - ③ Medium

① sender :-

A sender (or a group of senders) is responsible for initiating the diffusion process.

② Receiver :-

A Receiver (or a group of receivers) receives the diffusion information from the sender. Commonly, the no. of receivers is higher than the no. of senders.

③ Medium :-

This is the channel through which the diffusion information is sent from the sender to the receiver.

This can be TV, newspaper, social media (eg., a tweet on Twitter), social ties, air (in the case of a disease spreading process), etc.

- ③ from a network point of view : how is the diffusion process handed over? In fact, social relations play a significant role. They are the channels by which social contagion and persuasion are done. particularly, the structural positions of persons and their personal characteristics make some people more ready to adopt the innovation than others. N/W with different patterns of connection have different properties regarding how things are propagated, which have significant implications for interventions into,

for ex., rumor propagation.

- ④ A diffusion starts with an adopter (or a few no. of adopters) who spreads the innovation to others.
Innovation typically represent newness, it is not the same thing as invention, it is both a process and an outcome, and it involves discontinuous change.
- ⑤ Those who adopt early are often too innovative to be influential in local network. They contaminate their contacts who in turn contaminate their contacts and so on. The more people a person is linked to, the greater the chances that person will adopt the innovation. At a larger scale, and since communities are interlinked, it is very likely that an innovation jumps from one community to another via boundary spanners (or bridges) and start over diffusing again. It is a characteristic of social networks.
- ⑥ However, any diffusion process can be expedited, delayed, or even stopped if it is discovered that the product (e.g., a video, an audio, a book, etc) is faulty, and it should be fixed and then released again. this process is called an intervention.

③ Modeling Diffusion

Ex., community & cluster

IMP An introduction to the programming
screencast (coding 4 major ideas)

① The Base code

- ① coding the first Big idea - increasing the payoff
 - ① coding the second Big idea - key people
 - ① coding the third Big idea - impact of communities on cascades.
 - ① coding the fourth Big idea - cascades and clusters.
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